
Conjugate-Paired Fourier Gaussian Splats for Cryo-EM Reconstruction: A Feasibility Study

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Abstract

We study a direct Fourier-domain Gaussian representation for single-particle cryogenic electron microscopy. Complex Gaussian kernels are paired under conjugate reflection, enforcing the Hermitian symmetry of a real electrostatic potential. We derive the exact restriction of anisotropic kernels to projection planes, including plane-distance attenuation, contrast transfer, and translations. Fixed geometry yields a regularized linear least-squares problem, while a differentiable formulation supports geometry learning. We evaluate the stationary model on 24,576 experimental particle images from three EMPIAR accessions, using supplied consensus poses and 64×64 images. Conditional half-map FSC remains above 0.143 through the sampled limits of 7.87–16.08 Å. Mean shell-wise agreement with classical cryoDRGN backprojection is 0.984–0.995; these correlations measure agreement, not independent accuracy. Held-out image error is modestly lower than for a matched Fourier-voxel solver, but the reference implementation is slower. A separate low-frequency real-data experiment fits centers and anisotropic precisions. The results support feasibility at the evaluated bandwidth, without establishing ab initio inference, a speed advantage, or near-atomic heterogeneous reconstruction.

1. Introduction

Single-particle cryo-EM estimates molecular structure from many noisy projections. Neural reconstruction methods such as cryoDRGN represent conformationally varying volumes using coordinate networks (Zhong et al., 2021a). Their connection to neural fields motivates alternatives with explicit local basis functions. However, the cryo-EM projection is a linear integral, not the occlusion-dependent color

compositing used in ordinary neural radiance fields and graphics splatting.

Gaussian representations are already established in cryo-EM. E2GMM models heterogeneity using Gaussian mixtures (Chen & Ludtke, 2021); CryoSPIRE introduces hierarchical parts (Shekarforoush et al., 2025); CryoGS/CryoSplat and GEM exploit real-space Gaussian rendering (Chen & Ling, 2026; Qu et al., 2025). Merely substituting Gaussians for a neural network is therefore not a new contribution. Our specific proposal is to put complex kernels at nonzero *Fourier frequencies*, pair their coefficients by conjugation, and fit their plane restrictions. We do not claim that Fourier transforms themselves lose information: discretization, interpolation, and statistical estimation are the relevant approximations.

The scope is homogeneous reconstruction with supplied poses. This isolates the representation and its numerical conditioning. We distinguish the implemented stationary dictionary from proposed adaptive geometry and latent-variable extensions. The high-resolution adaptive and latent-variable extensions require additional validation. Reconstruction under known poses does not establish ab initio recovery, biological validity, or high-resolution heterogeneous reconstruction.

2. Related Work

CryoDRGN learns a latent-conditioned frequency-domain density (Zhong et al., 2021a); cryoDRGN2 jointly estimates poses and structure through multiresolution pose search (Zhong et al., 2021b). Classical Bayesian methods estimate voxel volumes and nuisance parameters (Scheres, 2012). These are different inference problems unless poses, particle selection, resolution, and compute budgets are controlled.

Real-space Gaussian mixtures have analytic Fourier transforms, but translating a real-space Gaussian induces a phase ramp on a Gaussian envelope centered at zero frequency. A Gaussian centered at a nonzero Fourier frequency instead corresponds to an oscillating, spatially localized function. Our proposed dictionary is thus a Gaussian-windowed spectral basis, closely related mathematically to Gabor functions. Its potential novelty lies in its particular cryo-EM inference

¹Fourier Cryo Splats research project. Correspondence to: Project repository <<https://github.com/yashpatel5400/fourier-cryo-splats>>.

and splatting construction, not the existence of Gaussian bases or the Fourier slice theorem. An exhaustive priority claim is outside this study.

3. Method

3.1. Image formation and conventions

Use the Fourier convention $F(k) = \int \rho(x) \exp(-2\pi i k^\top x) dx$. For an orthonormal plane embedding $E_i \in \mathbb{R}^{3 \times 2}$, detector frequency $q \in \mathbb{R}^2$, image translation t_i , and real even contrast transfer function C_i , the observation is

$$Y_i(q) = C_i(q) e^{-2\pi i q^\top t_i} F(E_i q) + \epsilon_i(q). \quad (1)$$

Frequencies must share units, such as inverse angstroms, and translations must use the corresponding length unit. Implementations using cycles per box must scale pixel sizes and translations consistently after downsampling. We assume the weak-phase, central-slice imaging model and neglect Ewald curvature, higher-order aberrations, and multiple scattering.

3.2. Conjugate-paired spectral kernels

Define $g(k; \mu, \Lambda) = \exp[-\frac{1}{2}(k - \mu)^\top \Lambda (k - \mu)]$ with positive-definite precision Λ . Our representation is

$$F_\theta(k) = \sum_{j=1}^K [c_j g(k; \mu_j, \Lambda_j) + \bar{c}_j g(k; -\mu_j, \Lambda_j)]. \quad (2)$$

Here $c_j \in \mathbb{C}$, $\mu_j \in \mathbb{R}^3$, and $\Lambda_j = L_j L_j^\top + \delta I$. A zero-centered component has a real coefficient and is included separately so that it is not counted twice. No opacity, transmittance, depth sorting, or view-dependent color appears.

Proposition 1. Equation (2) satisfies $F_\theta(-k) = \overline{F_\theta(k)}$, and hence its inverse Fourier transform is real.

Proof. $g(-k; \mu, \Lambda) = g(k; -\mu, \Lambda)$ is real. Reflection exchanges each pair, which is exactly complex conjugation of their sum. $\square$

The pair has the explicit inverse transform

$$\rho_j(x) = 2(2\pi)^{3/2} |\Lambda_j|^{-1/2} \times e^{-2\pi^2 x^\top \Lambda_j^{-1} x} \operatorname{Re}\{c_j e^{2\pi i \mu_j^\top x}\}. \quad (3)$$

It is a signed oscillatory function; positivity does not follow from Hermitian symmetry. A spatial support or nonnegativity penalty can be considered, but would alter the inference problem and must be reported explicitly.

3.3. Exact restriction to a plane

Let $A = E^\top \Lambda E$, $b = E^\top \Lambda \mu$, and $m = A^{-1}b$. Completing the square gives

$$g(Eq; \mu, \Lambda) = a \exp[-\frac{1}{2}(q - m)^\top A (q - m)], \quad (4)$$

$$a = \exp[-\frac{1}{2}(\mu^\top \Lambda \mu - b^\top A^{-1}b)].$$

The 2D covariance is A^{-1} : it is obtained by restricting *pre-cision*, not by projecting the 3D covariance as in a real-space Gaussian integral. The attenuation a is essential; a kernel far from the central plane should contribute negligibly. This formula permits plane culling and elliptical rasterization followed by CTF multiplication. If $|c_j|a_j \leq \tau_j$, omitting that component incurs at most τ_j absolute error per Fourier sample; errors add across omitted components. Culling must respect conjugate partners.

3.4. Inference

For known poses and geometry, stack independent real and imaginary observations into y , let B_η be the real linear operator generated by Equation (1), and solve

$$\min_a \frac{1}{2} \|W^{1/2}(B_\eta a - y)\|_2^2 + \frac{\lambda}{2} \|Da\|_2^2. \quad (5)$$

W contains inverse noise variances, D may penalize high-frequency coefficients, and a stores real and imaginary coefficient parts with the imaginary DC coefficient fixed to zero. One representative per Friedel pair is sufficient; using both simply duplicates information if weights are adjusted consistently. CTF zeros are handled by the likelihood, not unstable division.

The normal equations are

$$(B_\eta^\top W B_\eta + \lambda D^\top D)a = B_\eta^\top W y. \quad (6)$$

Matrix-free conjugate gradients or stochastic optimization can exploit local kernel support. A useful reference implementation keeps centers on a regular frequency lattice, uses a shared isotropic width, and fits all complex coefficients. This stationary case is a Gaussian radial-basis discretization, not evidence that learned sparse geometry outperforms conventional gridding.

For adaptive kernels, define the reduced objective $\mathcal{J}(\eta) = \min_a \mathcal{L}(a, \eta)$. At an exact inner optimum, its gradient is the partial derivative of $\mathcal{L}$ with respect to η (the envelope theorem). Inner solves must be converged sufficiently; otherwise ignoring their dependence on η biases the outer gradient. Center displacement, eigenvalue bounds, frequency marching, and a coefficient penalty stabilize the otherwise ill-conditioned basis. Split or prune kernels only using training residuals and coverage, with a separate validation set for model selection.

3.5. Cost and limitations

Dense evaluation costs $O(KM)$ for M Fourier samples. Culling yields cost proportional to the number of sample-kernel intersections, but a speedup requires bounded overlap and an efficient implementation; it is not guaranteed by the representation. Finite neighborhood truncation approximates infinite Gaussian support and should be checked against a larger neighborhood. Narrow kernels poorly couple nearby slices; broad kernels can make coefficient recovery ill-conditioned. Missing viewing directions and CTF coverage cannot be repaired by smoothness alone.

4. Experimental Protocol

4.1. Data and matched baselines

We select EMPIAR-10028 (Pf80S ribosome), EMPIAR-10049 (RAG complex), and EMPIAR-10076 (assembling 50S subunit) (EMPIAR, 2015a;b; 2016), using deposited experimental particle images and public cryoDRGN poses/CTFs. These are extracted images, not a fresh movie-motion-correction and particle-picking pipeline. We record particle counts, selection indices, original and working pixel sizes, Fourier crop size, signs, seeds, and downloaded byte-range hashes. Different dataset accessions are not replaced by three synthetic stacks.

For a controlled known-pose comparison, we fit a classical Fourier voxel representation with trilinear interpolation and the same objective, particle partitions, and frequency support. We additionally run cryoDRGN’s classical CTF-weighted backprojection, with its version and invocation recorded. Ab initio comparisons require independent pose estimation and are a distinct experiment.

4.2. FSC and safeguards

For two Fourier volumes and shell S_r , we compute

$$\text{FSC}(r) = \frac{\text{Re} \sum_{k \in S_r} F_1(k) \bar{F}_2(k)}{\sqrt{\sum_{k \in S_r} |F_1(k)|^2 \sum_{k \in S_r} |F_2(k)|^2}}. \quad (7)$$

Independently fitted particle halves with consensus poses give *conditional half-map FSC*; they are not fully independent gold-standard reconstructions (Scheres & Chen, 2012; Chen et al., 2013). Method-baseline FSC measures agreement, not resolution. We use unmasked FSC as the primary transparent diagnostic; common masks can inflate correlation and require noise-substitution correction. We report the first sustained crossing of 0.143 and the sampled frequency limit, censoring curves with no crossing. Alignment, sharpening, masks, and hyperparameters are not optimized on the reported FSC. Held-out image prediction and noise controls complement FSC, because shared bias can inflate agreement.

Table 1. Experimental inputs. All runs use 8,192 particles per accession. Pixel sizes are in Å/pixel; “raw” denotes deposited extracted images.

EMPIAR	Raw box	Raw pixel	Working pixel
10028	360	1.34	7.5375
10049	192	1.23	3.6900
10076	320	1.31	6.5500

5. Results

5.1. Executed study

We downloaded real deposited particle bytes for 8,192 particles from each of three distinct accessions, totaling 24,576 experimental images. Selection used randomly permuted contiguous blocks of 64 in the source order, applying the published cryoDRGN particle filter when present. This is a reproducible block sample, not an independent uniform sample or the entire deposited dataset. Original selected images were Fourier-cropped to 64×64 while preserving the physical field of view. Source byte ranges, SHA-256 hashes, particle indices, and the metadata repository commit are included in the reproducibility package. No detector-movie motion correction or new particle picking was performed. One EMPIAR-10028 source file was excluded because its header declared 113 particles but the served file contained only 4 KB. The 32 affected selected particles were replaced by continuing the same seeded selection order; the exclusion and identity-preserving recovery are documented in the repository.

For each accession, a fixed permutation (seed 42 for observations; 43 for partitioning) reserves 409 images for validation and 409 for test. The remaining 7,374 images are split into two disjoint sets of 3,687. We use all 1,410 nonredundant Fourier samples inside radius 30, excluding DC. Images are divided by their background standard deviation and given the documented cryoDRGN sign convention. The standard linear image-window taper from 0.85 to 0.99 of the half-box is used for the final experiment. Windowing introduces Fourier noise correlations; our diagonal least-squares objective is therefore an approximation, shared by the matched methods. The final map is the arithmetic mean of the two fitted maps.

The stationary Gaussian dictionary has centers on a 65^3 lattice, conjugate tied across the origin, width 0.5 Fourier bins, and radial truncation at 2 bins. The implementation allocates 274,625 real scalar coefficient slots, including unsupported locations driven to zero. It does not learn these centers or widths in the main three-dataset experiment. Preconditioned conjugate gradients solve the real and imaginary systems separately with relative tolerance 10^{-5} and at most 80 iterations. The ridge coefficient is $\lambda = \alpha$ times the median positive diagonal of the corresponding data normal matrix.

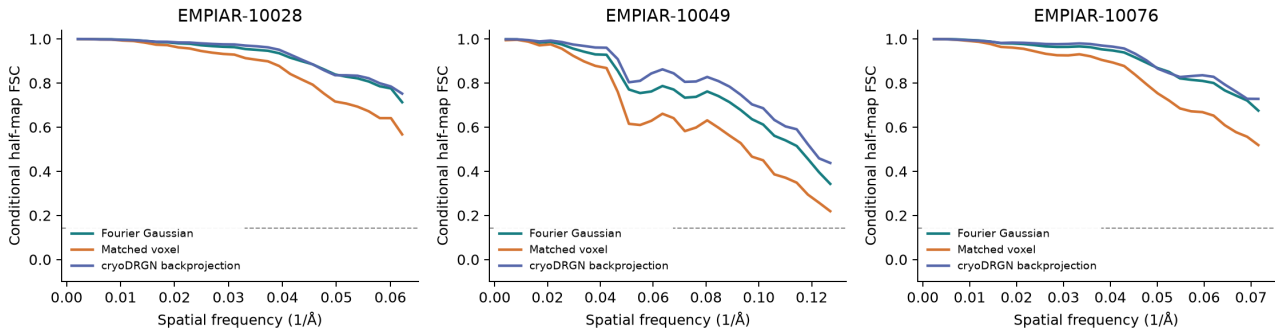


Figure 1. Conditional half-map FSC on three experimental particle selections. Poses are supplied from public consensus refinements, so these are not fully independent gold-standard reconstructions. The dashed line is 0.143.

Table 2. RAG validation NMSE used for selecting regularization. Lower is better.

α	Gaussian	Matched voxel
0.1	0.930710	0.930861
1	0.929539	0.930356
10	0.940827	0.947849

The matched baseline replaces the Gaussian basis by trilinear Fourier voxels while preserving data, frequency samples, objective, partitions and solver. A second baseline invokes cryoDRGN 4.3.1 `backproject_voxel`, with CTF multiplication, regularization weight 1, the same particle halves and the software’s standard image window. That external baseline uses its native backprojection rule and circular frequency support; FSC comparisons restrict all maps to the same radius-30 sphere. It is a classical fixed-pose command, not a neural or ab initio cryoDRGN experiment.

5.2. Regularization selected by prediction

We selected α from $\{0.1, 1, 10\}$ using RAG validation-image error, independently for each representation. Both selected $\alpha = 1$, which was then frozen for the other accessions. Neither the reported FSC curves nor test error selected this value. Table 2 shows why a high FSC alone is insufficient: the strongest regularizer can preserve high correlation while underpredicting the observed signal.

5.3. Reconstruction agreement and FSC

Figure 1 reports unmasked conditional half-map FSC. No spatial map mask, sharpening, or test-derived alignment is used. Table 3 distinguishes a measured threshold crossing from right-censoring at the last sampled frequency. A censored value is a bound at this experiment’s sampling limit, not a measurement of a finer resolution. Figure 2 shows representative map sections.

Table 3. Conditional half-map FSC at 0.143, in Å. A $\leq$ sign means the curve never crosses within the sampled band. BP is cryoDRGN’s classical backprojection. No result in this table is an ab initio resolution estimate.

EMPIAR	Gaussian	Voxel	BP
10028	≤ 16.08	≤ 16.08	≤ 16.08
10049	≤ 7.87	≤ 7.87	≤ 7.87
10076	≤ 13.97	≤ 13.97	≤ 13.97

Table 4. Mean shell-wise cross-method FSC over shells 1–30. These are agreement statistics between methods sharing images and poses, not independent accuracy or resolution measurements.

EMPIAR	Gaussian-voxel	Gaussian-BP
10028	0.9920	0.9941
10049	0.9806	0.9835
10076	0.9909	0.9949

5.4. Held-out prediction and cost

Table 5 reports normalized mean squared error on the 409 test images excluded from fitting and regularizer selection. Values close to one are expected for noisy particle images, since the denominator includes measurement noise. A small improvement should not be interpreted as a large structural advance. Supplemental particle-bootstrap intervals are conditional on the supplied poses and do not account for correlations within micrographs.

The main runs use CPU sparse linear algebra on an Apple M4 Pro with 24 GiB unified memory. The reference Gaussian implementation is slower than the trilinear baseline, with much of its cost in sparse kernel construction. These runs do not establish a speed or memory advantage. Concurrent downloads and operating-system activity also limit the precision of timing comparisons.

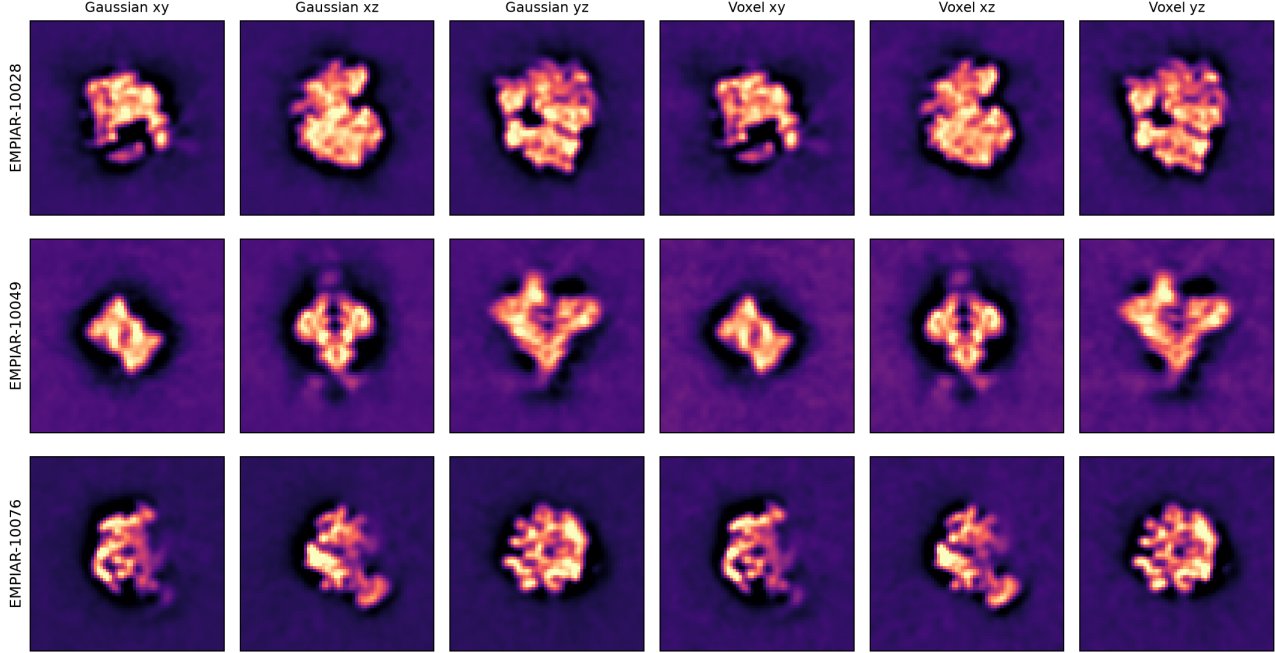


Figure 2. Central sections through the reconstructed mean maps. Rows are EMPIAR-10028, 10049, and 10076; columns show three orthogonal sections of Gaussian and matched voxel maps. For visualization only, each volume is smoothed by one voxel and displayed on its own intensity scale. FSC uses the unsmoothed maps.

Table 5. Held-out test NMSE and elapsed reconstruction time. G/V denote Gaussian and matched voxel. Timings include sparse operator construction, solves, map export, and held-out prediction; they exclude downloads and initial data loading.

EMPIAR	G NMSE	V NMSE	G (s)	V (s)
10028	0.83313	0.83939	102.3	13.0
10049	0.93023	0.93103	108.9	13.9
10076	0.87503	0.87909	96.0	13.1

5.5. Controls and a nonlinear feasibility run

An out-of-dictionary synthetic phantom is generated from displaced real-space Gaussians. The low-shell mean FSC of the recovered Fourier Gaussian model is

0.9933, versus 0.9836 for the voxel baseline.

This is an implementation check, not an experimental-data accuracy result. Analytic plane restriction, Hermitian symmetry, sparse adjoints, CTF parity, FSC behavior, inverse-transform reality and nonlinear geometry gradients also pass numerical checks. Increasing the Gaussian evaluation cutoff from 2 to 3 bins at fixed fitted coefficients produces the relative complex prediction errors recorded with the released validation outputs (maximum 0.00064 across datasets); this checks rendering truncation, not the effect of refitting with a different basis.

A phase-randomized control is fitted independently on each accession at 32×32 , with 256 Fourier samples per image. This preserves Fourier magnitudes while destroying coherent phase information. We report its entire FSC curves, rather than a single favorable shell. The mean absolute Gaussian-control FSC over shells 4–14 is 0.028, 0.010, 0.043 for the three accessions, respectively. None of these controls is a substitute for independent half-set pose refinement.

The general differentiable representation is separately tested on low-frequency RAG observations using a 16×16 grid, 709 conjugate pairs, and joint optimization of complex coefficients, centers and full anisotropic precisions. This MPS run uses 7,680 training images, 512 validation images, 1,500 Adam steps, bounded center displacements, and bounded precision parameters. The initial and best validation NMSE are 1.0000 and 0.7726, respectively, with 36.3 seconds of optimization. Its partition and band differ from the main benchmark. This demonstrates nonlinear fitting feasibility only; it does not establish improved resolution, sparse adaptive efficiency, or superiority to the fixed dictionary.

6. Discussion

The proposed spectral representation yields an exact central-plane restriction and a convex coefficient-fitting subproblem. Its practical value depends on conditioning, truncation error,

predictive performance, and measured cost. Fixed-pose, downsampled consensus reconstructions cannot resolve the full heterogeneous distributions modeled by cryoDRGN, and favorable correlation at a coarse frequency limit would not establish near-atomic accuracy. High-resolution adaptive anisotropy, independent pose inference, and heterogeneous extensions require further experiments.

Code and Data Availability

The code, source-index manifests, complete FSC curves, software environment, and manuscript are available at <https://github.com/yashpatel5400/fourier-cryo-splats>. Reconstructed half maps and coefficient checkpoints are distributed as release artifacts. The original experimental images remain available from the cited EMPIAR accessions; background papers are linked rather than redistributed.

Impact Statement

This work studies a computational representation for structural biology. Reconstructions may contain artifacts and should not be treated as validated atomic structures or used for biological inference without independent validation. We release provenance, failed runs where informative, and evaluation code to reduce the risk of overstated resolution claims.

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